3.6.2 Relative Gain Array (RGA)

The relative gain array (RGA) of a non-singular square matrix G is a square matrix defined as

$$\mathrm{RGA}(G) = \Lambda(G) \triangleq G \times (G^{-1})^T \tag{3.69}$$

where $\times$ denotes element-by-element multiplication (the Hadamard or Schur product). For a 2×2 matrix with elements g_{ij} the RGA is

$$\Lambda(G) = \begin{bmatrix} \lambda_{11} & \lambda_{12} \\ \lambda_{21} & \lambda_{22} \end{bmatrix} = \begin{bmatrix} \lambda_{11} & 1-\lambda_{11} \\ 1-\lambda_{11} & \lambda_{11} \end{bmatrix}; \quad \lambda_{11} = \frac{1}{1 - \frac{g_{12}g_{21}}{g_{11}g_{22}}} \tag{3.70}$$

Bristol (1966) originally introduced the RGA as a steady-state measure of interactions for decentralized control. Unfortunately, based on the original definition, many people have dismissed the RGA as being "only meaningful at $\omega = 0$". To the contrary, in most cases it is the value of the RGA at frequencies close to crossover which is most important.

The RGA has a number of interesting *algebraic properties*, of which the most important are (see Appendix A.4 for more details):

1. It is independent of input and output scaling.
2. Its rows and columns sum to one.
3. The sum-norm of the RGA, $\|\Lambda\|_{\mathrm{sum}}$, is very close to the minimized condition number γ^*; see (A.78). This means that plants with large RGA-elements are always ill-conditioned (with a large value of $\gamma(G)$), but the reverse may not hold (i.e. a plant with a large $\gamma(G)$ may have small RGA-elements).
4. A relative change in an element of G equal to the negative inverse of its corresponding RGA-element yields singularity.
5. The RGA is the identity matrix if G is upper or lower triangular.

From the last property it follows that the RGA (or more precisely $\Lambda - I$) provides a measure of *two-way interaction*. The definition of the RGA may be generalized to non-square matrices by using the pseudo inverse; see Appendix A.4.2.

In addition to the algebraic properties listed above, the RGA has a surprising number of useful *control properties*:

1. The RGA is a good indicator of sensitivity to uncertainty:
   (a) *Uncertainty in the input channels (diagonal input uncertainty)*. Plants with large RGA-elements around the crossover frequency are fundamentally difficult to control because of sensitivity to input uncertainty (e.g. caused by uncertain or neglected actuator dynamics). In particular, decouplers or other inverse-based controllers should not be used for plants with large RGA-elements (see page 244).
   (b) *Element uncertainty.* As implied by algebraic property no. 4 above, large RGA-elements imply sensitivity to element-by-element uncertainty. However, this

kind of uncertainty may not occur in practice due to physical couplings between the transfer function elements. Therefore, diagonal input uncertainty (which is always present) is usually of more concern for plants with large RGA-elements.

2. *RGA and RHP-zeros.* If the sign of an RGA-element changes from $s = 0$ to $s = \infty$, then there is a RHP-zero in G or in some subsystem of G (see Theorem 10.5).
3. *Non-square plants*. Extra inputs: If the sum of the elements in a column of RGA is small ($\ll 1$), then one may consider deleting the corresponding input. Extra outputs: If all elements in a row of RGA are small ($\ll 1$), then the corresponding output cannot be controlled (see Section 10.4).
4. *Diagonal dominance.* The RGA can be used to measure diagonal dominance, by the simple quantity

$$\text{RGA-number} = \|\Lambda(G) - I\|_{\text{sum}} \tag{3.71}$$

For decentralized control we prefer pairings for which the RGA-number at crossover frequencies is close to 1 (see pairing rule 1 on page 435). Similarly, for certain multivariable design methods, shaping, it is simpler to choose the weights and shape the plant if we first rearrange the inputs and outputs to make the plant diagonally dominant with a small RGA-number.

5. *RGA and decentralized control.*

   (a) *Integrity:* For stable plants avoid input-output pairing on negative steady-state RGA-elements. Otherwise, if the sub-controllers are designed independently each with integral action, then the interactions will cause instability either when all of the loops are closed, or when the loop corresponding to the negative relative gain becomes inactive (e.g. because of saturation) (see Theorem 10.4 page 439). Interestingly, this is the only use of the RGA directly related to Bristol's original definition.

   (b) *Stability:* Prefer pairings corresponding to an RGA-number close to 0 at crossover frequencies (see page 435).

Remark. An iterative evaluation of the RGA, $\Lambda^2(G) = \Lambda(\Lambda(G))$ etc., has in applications proved to be useful for choosing pairings for large systems. Wolff (1994) found numerically that

$$\Lambda^\infty \triangleq \lim_{k\to\infty} \Lambda^k(G) \tag{3.72}$$

is a permuted identity matrix (with the exception of "borderline" cases, the result is proved for a positive definite Hermitian matrix G by Johnson and Shapiro (1986)). Typically, Λ^k approaches Λ^∞ for k between 4 and 8. This permuted identity matrix may then be used as a candidate pairing choice. For example, for $G = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$ we get $\Lambda = \begin{bmatrix} 0.33 & 0.67 \\ 0.67 & 0.33 \end{bmatrix}$, $\Lambda^2 = \begin{bmatrix} -0.33 & 1.33 \\ 1.33 & -0.33 \end{bmatrix}$, $\Lambda^3 = \begin{bmatrix} -0.07 & 1.07 \\ 1.07 & -0.07 \end{bmatrix}$ and $\Lambda^4 = \begin{bmatrix} 0.00 & 1.00 \\ 1.00 & 0.00 \end{bmatrix}$, which indicates that the off-diagonal pairing should be considered. Note that Λ^∞ may sometimes "recommend" a pairing on negative RGA-elements, even if a positive pairing is possible.